

Instructor: Dr. Rajesh Kumble Nayak

3:00 PM, 25 September, 2024.

Duration 90 Min.



- Read each question carefully.
- Answer all questions.
- Ensure all parts of the answer for a given question are in the same place.
- Write neatly, and do all rough work on the answer sheet.
- Do not write on the question paper.
- Keep answers concise and to the point.

Good luck!

Q - 1(20 Marks)In a generalised coordinate system $\{\sigma, \tau\}$ the Lagrangian is given by,

$$\mathcal{L} = \frac{m}{2} \left(\frac{1}{\sigma^2 + \tau^2} \right) [\dot{\sigma}^2 + \dot{\tau}^2]$$

1. Find the Hamiltonian for the system
2. Find the Hamiltonian Equation of motion.

Q - 2(20 Marks)In a generalised coordinate system $\{q_1, q_2\}$ the Lagrangian is given by,

$$\mathcal{L} = \dot{q}_1^2 + \dot{q}_2^2 + k_1 q_1^2 + k_2 q_1 \dot{q}_2,$$

1. Find the generalised momenta $\{p_1, p_2\}$.
2. Express the generalised velocities $\{\dot{q}_1, \dot{q}_2\}$ in terms of generalised momenta $\{p_1, p_2\}$.
3. Find the Hamiltonian for the system. It is not necessary to simplify the Hamiltonian to its simplest form.

Q - 3(5 Marks)For a Hamiltonian system, the Hamiltonian $H(q_i, p_i, t)$ satisfies the Hamiltonian equations of motion. Show that if $\frac{\partial H}{\partial t} = 0$, then $\frac{dH}{dt} = 0$ or H is constant of motion.**Q - 4(5 Marks)**

Lagrangian for a system is given by

$$\mathcal{L} = \frac{m}{2} (\dot{x}^2 + \dot{y}^2) + q (\dot{x}A_x + \dot{y}A_y) - q\Phi(x, y)$$

Here, $\vec{A} = (A_x, A_y)$. Find the Hamiltonian.